Eric Xu

Applied math M.S. | *Machine learning for rare-event forecasting, scientific computing, GPU inference*
exu23@ucsc.edu | ericxu1070.github.io | github.com/ericxu1070 | linkedin.com/in/exu06

Education

University of California, Santa Cruz — Santa Cruz, CA
M.S. in Scientific Computing and Applied Mathematics (Plan I thesis) — *Expected June 2027*
B.S. in Applied Mathematics, Highest Honors in the Major | GPA 3.77/4.00 | Dean's Honors (4 quarters) — *2024 – 2026*

- **Relevant coursework:** Scientific Machine Learning (grad), Applied Dynamical Systems (grad), Numerical Linear Algebra (grad, in progress), GPU Programming, Probability and Statistics, Classical and Bayesian Inference, Financial Mathematics

Research Experience

TACS Lab, UC Santa Cruz — Santa Cruz, CA
Researcher, B.S. and M.S. (Advisor: Prof. Ashesh Chattopadhyay) — *Sep 2025 – Present*

- Adapt AI+RES rare-event sampling (Lancelin et al., 2025) to subseasonal forecasting: AI weather-model ensembles (FourCastNet 3, GenCast) are cloned and pruned by score to sample heat waves and cold outbreaks 3 to 4 weeks ahead.
- Built the batch pipeline (JAX, PyTorch; Slurm on H100 nodes, PBS on NCAR Derecho) that launches ensembles weeks before past US extremes and checks against ERA5 (0.25°) whether they reach the observed event.

Vayuh.ai
ML Researcher (part-time) — *Jul 2026 – Present*

- Made GenCast's 0.25° JAX executable small enough for the persistent compile cache and added persistent GPU workers, so each forecast case compiles the model once instead of at every leg.
- Benchmarked AI weather models (GenCast, FourCastNet 3, SFNO) as ensemble members on H100 nodes and moved the ensemble to FourCastNet 3, cutting GPU-hours per forecast case.

Projects

Council of Sumer | *Python, FastAPI, AGNTCY SLIM, OpenTelemetry* — Oct 2026

- Built a council of five LLM agents with conflicting goals (hackathon, team of 3): they run as separate processes, debate over AGNTCY SLIM, and vote under runoff and human tie-break rules to govern the city in Hamurabi.
- Co-wrote the deliberation engine and benchmark: over 16 seeded games the council survived 11 of 15 completed reigns vs 9 of 16 for a single LLM, and starved 27% fewer people.

Cross-Species Transfer Learning with DeepSEA | *PyTorch, scikit-learn* — Jun 2026

- Fine-tuned the human-trained DeepSEA CNN on mouse ENCODE DNA windows. It beat training from scratch on 115 of 133 chromatin features (median AUROC 0.877 vs 0.865) and converged in fewer epochs.
- Showed that zero-shot transfer to mouse scores near chance (median AUROC 0.50), so the model needs fine-tuning.

4D-Var Data Assimilation with a Learned Inverse Operator | *PyTorch, L-BFGS* — Apr – May 2026

- AM 170B capstone (team of 5): reimplemented Frerix et al. (ICML 2021) in PyTorch and extended it with the full 4D-Var cost, Gaussian observation noise at three levels, and sliding-window cycling, on Lorenz-96 and 2D Kolmogorov flow.
- On Lorenz-96, learned-inverse initialization with hybrid optimization gave the lowest analysis RMSE at every noise level.

Teaching Experience

Baskin School of Engineering, UC Santa Cruz — Santa Cruz, CA
Teaching Assistant: Ordinary Differential Equations, Differential Calculus for Economics — *Sep 2026 – Present*

- Lead four weekly discussion sections (two per course) and office hours, and grade assignments.

Tutor (3 quarters) and Reader (1 quarter): Linear Algebra, Discrete Mathematics, Calculus II — *Sep 2025 – Jun 2026*

Skills

Languages: Python (primary), C++ (working knowledge), MATLAB, R, Bash; familiar: RISC-V assembly
ML and Scientific Computing: PyTorch, JAX, NumPy, scikit-learn; familiar: Triton (GPU kernels)
HPC and Tools: Slurm, PBS, NCAR Derecho, multi-GPU H100 nodes, Linux, Git, SSH
Mathematics: Linear algebra, ODEs and PDEs, dynamical systems, numerical methods, probability and statistics